Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 5 [2011 P2]

Test Date: 20th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCE-2011-P2-Model

Time: 3:00 **IMPORTANT INSTRUCTIONS** **Max Marks: 240**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 9 – 12)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 13 – 18)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 19 – 20)	Matrix Matching Type	8	0	2	16
Total				20	80

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 21 – 28)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 29 – 32)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 33 – 38)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 39 – 40)	Matrix Matching Type	8	0	2	16
Total				20	80

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 41 – 48)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 49 – 52)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 53 – 58)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 59 – 60)	Matrix Matching Type	8	0	2	16
Total				20	80

CHEMISTRY: **Max.Marks: 80**

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

1. A saturated solution of AgBr. K_{sp} of AgBr is 12×10^{-14}. *If* 10^{-7} moles of $AgNO_3$ are added to 1 litre of this solution then the conductivity of this solution in terms of $10^{-7}\,Sm^{-1}$ units will be $\left[\text{Given } \lambda^{o}_{(Ag^{+})}=4\times10^{-3}\,Sm^{2}mol^{-1}\,\lambda^{o}_{(Br^{-})}=6\times10^{-2}\,Sm^{2}mol^{-1},\lambda^{o}_{(NO_3^{-})}=5\times10^{-3}\,Sm^{2}mol^{-1}\right]$

A) 39 **B)** 55 **C)** 15 **D)** 41

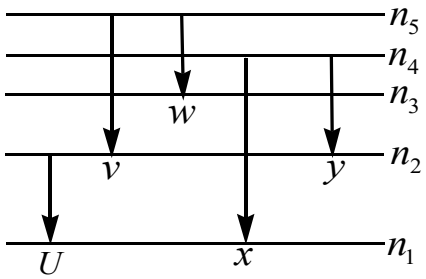
2. In a hypothetical reaction $A(aq)\rightleftharpoons 2B(aq)+C(aq)$ (1st order decomposition)

'A' is optically active (dextro-rotatory) while 'B' and 'C' are optically inactive but 'B' takes part in a titration reaction (fast reaction) with H_2O_2. Hence the progress of reaction can be monitored either by measuring rotation of plane polarised light or by measuring volume of H_2O_2 consumed in the titration. In an experiment, the optical rotation was found to be $\theta=30^{o}$ at t = 20 min. and $\theta=15^{o}$ at t = 50 min. from start of the reaction. If the progress would have been monitored by titration method, then the volume of H_2O_2 consumed at t = 30 min. (from start) is 30 ml then the volume of H_2O_2 consumed at t = 90 min. will be:

A) 60ml **B)** 45ml **C)** 52.5 ml **D)** 90 ml

3. For a hypothetical hydrogen like atom which follows Bohr's model, some spectral lines were observed as shown. If it is known that line 'U' belongs to the visible region, then the lines possibly belonging to ultra violet region will be (n_1 is not necessarily ground state)

[Assume for this atom, no spectral series shows overlap with other series in the emission spectrum]



A) y and v **B)** v only **C)** w only **D)** x only

4. Which of the following are correct statements

A) Spontaneous adsorption of gases on solid surface is an exothermic process and entropy decreases during adsorption

B) Formation of micelles takes place when temperature is below Kraft Temperature (T_K) and concentration is above critical micelle concentration (CMC)

C) A colloid of $Fe(OH)_3$ is prepared by adding a little excess (than required to completely precipitate of Fe^{3+} ions as $Fe(OH)_3$) of NaOH to $FeCl_3$ solution then the particles of this sol moves towards cathode during electrophoresis.

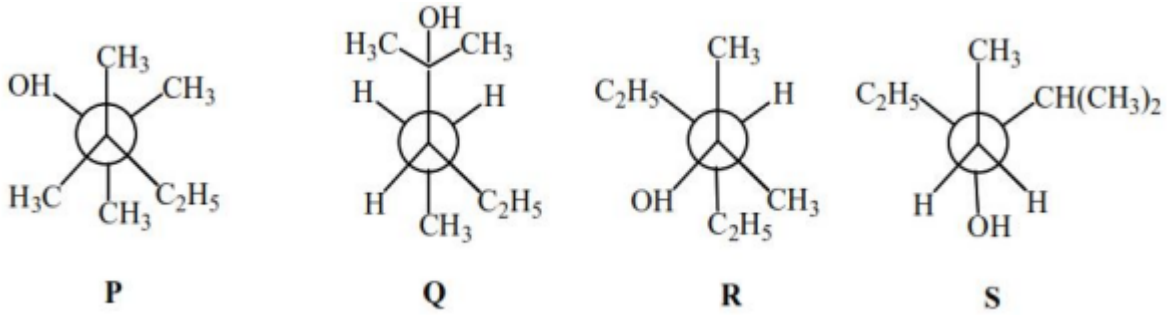
D) According to Hardy-Schulze rules the coagulation (flocculating) value of Fe^{3+} ion will be more than Ba^{2+} *or* Na^{+}

5. Which of the following option is incorrectly matched

A) S_3O_9 ; contains S-S linkage. **B)** $S_2O_6^{2-}$; contains S-S linkage.

C) $(HPO_3)_3$; contains P-O-P linkage **D)** $S_2O_8^{2-}$; contains O-O linkage

6. Newman projections P, Q, R and S are shown below:



P **Q** **R** **S**

Which one of the following options represents identical molecules

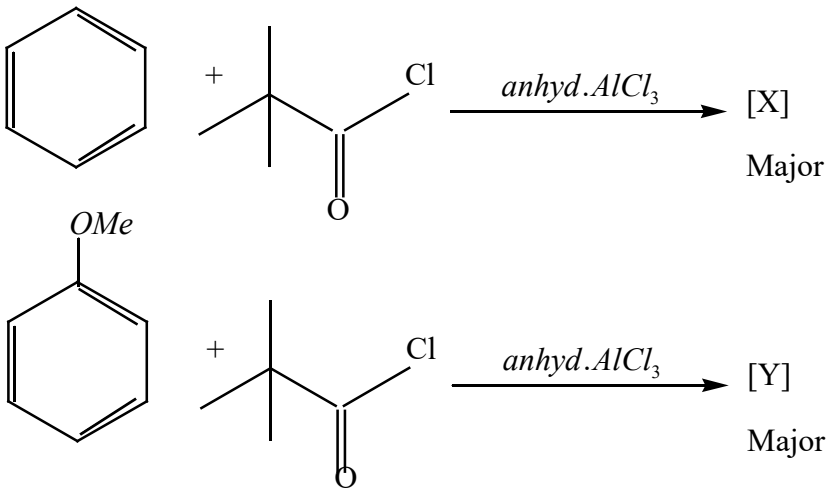
A) P and Q **B)** Q and S **C)** Q and R **D)** R and S

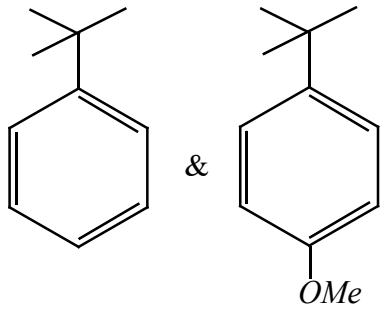
7. Which of the following is more basic than N, N-dimethyl aniline?

A) Phenylmethanamine **B)** Acetanilide

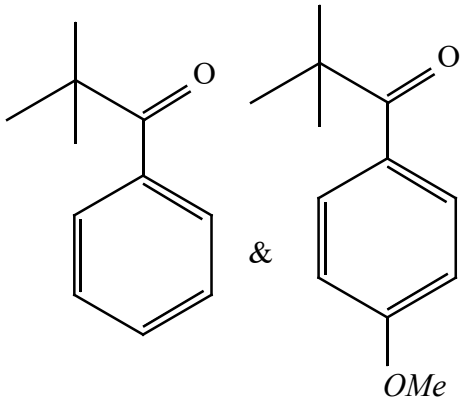
C) Aniline **D)** Benzamide

8. Compounds [X] and [Y] in the following reactions respectively are

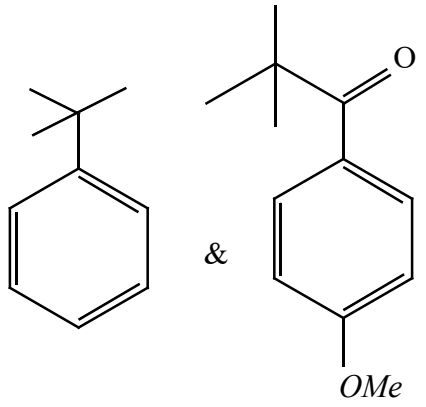




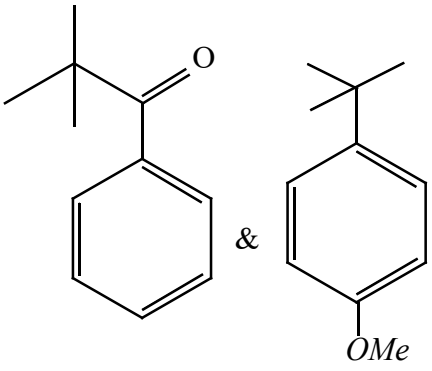
A)



B)



C)



D)

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

9. Two liquids A and B form an ideal solution. The solution has a vapour pressure of 700 torr at $80^{o}C$. It is distilled till 2/3rd of the solution is collected as condensate. The composition of the condensate is $X_A^{'} = 0.75$ and that of the residue is $X_A^{''} = 0.30$. If the vapor pressure of the residue at $80^{o}C$ is 600 Torr, then which of the following is/are true

A) The composition of the original liquid was $X_A = 0.6$

B) The composition of the original liquid was $X_B = 0.8$

C) $P_A^0 = \frac{2500}{3} Torr$

D) $P_B^0 = 500\,Torr$

10. Identify the correct options

A) Order of bond angle $NH_4^+ > NH_3 > NH_2^-$

B) $(CH_3)_3 B$ is a trigonal planar molecule (not considering the H-atoms on 'C')

C) ClO_2 reacts with ozone forming a higher oxide of chlorine in which the oxidation of state of chlorine is +7

D) S_8 molecule has a total of 16 non-bonded electrons on all the 'S' atoms after bonding.

11. For Mn^{+3} pairing energy is $28000\,cm^{-1}$, $\Delta_0\ for\left[Mn(CN)_6\right]^{3-}$ is $38500cm^{-1}$ then which of the following is/are correct

A) In the complex the central metal ion has configuration $t_{2g}^4 e_g^0$

B) Complex will be low spin complex

C) Net CFSE (including pairing energy) $= -33600cm^{-1}$

D) The complex is paramagnetic

12 **A.** $Ca_2B_6O_{11} + Na_2CO_3 \xrightarrow{\Delta} [X] + CaCO_3 + NaBO_2$ (unbalanced equation)

Correct choice(s) for [X] is/are:

A) structure of anion of crystalline [X] has one boron atom sp^3 hybridized and other three boron atoms sp^2 hybridized

B) X upon strong heating gives a glassy bead consisting of boric anhydride and $NaBO_2$ aqueous $NaBO_2$ reacts with hydrogen peroxide in yields a boran compound which is used as brightner in laundry detergents.

C) hydrolysis of [X] with HCl or H_2SO_4 yields a compound which on reaction with HF gives fluoroboric acid.

D) [X] on heating with copper salts in oxidizing flame gives blue coloured bead after cooling.

SECTION –III

(INTEGER ANSWER TYPE)

This section contains **6 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS

13. How many of the following statements are correct?

(a) BeO is amphoteric in nature.

(b) $LiHCO_3$ is not found in solid state.

(c) K_2O_2 is paramagnetic but KO_2 is diamagnetic.

(d) White phosphorous react with caustic soda gives phosphine gas.

(e) $AlCl_3$ is soluble in excess of NaOH and form sodium meta aluminate.

(f) Anhydrous silver(I)carbonate upon heating gives metallic silver.

(g) Lithium fluoride is highly soluble in water.

(h) Hydrated magnesium chloride on heating in dry air gives anhydrous $MgCl_2$.

14. A black compound (A) in solid state is fused with KOH and $KClO_3$. The aqueous extract of fused mass is a green coloured solution due to the compound(B). On passing CO_2 gas through it produces a pink coloured solution due to the compound (C) along with some black insoluble mass of (A). The pink-coloured solution is decolorized by Fe^{2+} in acidic medium. The oxidation state of central metal in (C) is_____.

15. An alcoholic solution of dimethylglyoxime is added to an aqueous solution of nickel(II) chloride. Slow addition of ammonium hydroxide led to the precipitation of a bright-red coloured metal complex. The number of hydrogen bonds present in the coloured metal complex structure is_____.

16. How many statements are correct?

(i) SO_2 also gives lime water test like CO_2 gas .

(ii) In the test of NO_3^- , brown ring is formed due to the charge transfer complex $\left[Fe(NO)\right]SO_4$

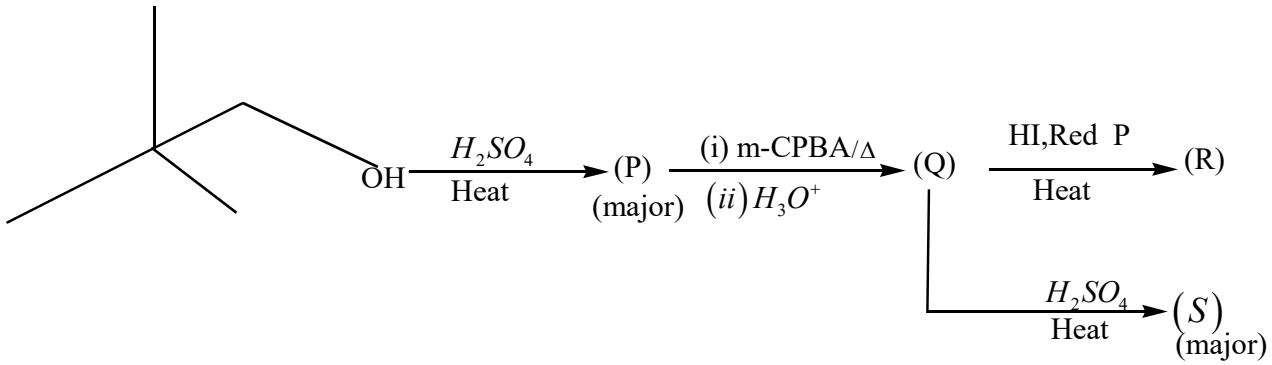
(iii) Aqueous solution of Ag^+ ions gives brick red/red precipitate with K_2CrO_4.

(iv) Lead salts turn black on prolonged exposure in a chemical laboratory.

(v) In analysis of group II[nd] cations, H_2S gas is passed in presence of hydrochloric acid to enhance the concentration of S^{2-} ions.

(vi) Aqueous solution of Cu^{2+} ions forms a green complex with $K_4[Fe(CN)_6]$.

17. For the given reaction



Correct statements is / are about products:

i) Number of all possible monobromo products for (R) are 6 (including stereo)

ii) (P) Show positive bromine water test

iii) Q) Gives methane gas with CH_3MgBr

iv) (Q) Can react with Na metal

v) (S) gives positive 2,4 D N P test

vi) (P) Upon reductive ozolysis gives acetone and acetaldehyde

vii) (S) gives negative Tollen's test

18. An organic compound of formula $C_6H_{14}O(A)$,gives white turbidity in Lucas test in 5 to 7 minutes and also gives yellow precipitate with I_2 in presence of aqueous NaOH. If the total no. of isomers possible for A are x, optically active isomers possible for A are y and fractions of A after fractional distillation are z then find the value of $\frac{x+y}{z}$.

SECTION –IV

(MATRIX-MATCH TYPE)

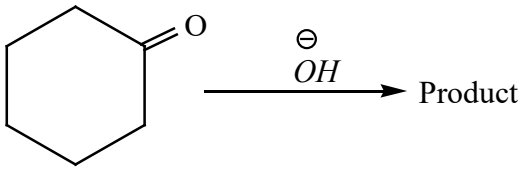
This section contains 2 questions. Each question has **for statements** (A,B,C and D) given in **Column I** and **five statements** (p,q,r,s, and t) in **Column II.** Any given statement in **Column I** can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statements B, darken the bubbles corresponding to q and r in the ORS.

19. Match each coordination compound in List-I with an appropriate pair of characteristics from List-II and select the correct answer using the code given below the lists.

$\{en = H_2NCH_2CH_2NH_2; atomic\ numbers: Ti = 22; Cr = 24; Co = 27; Pt = 78\}$

List-I		List-II	
a	$\left[\left(Cr\left(NH_3\right)_4 Cl_2\right)Cl\right]$	p	Paramagnetic
b	$\left[Ti\left(H_2O\right)_5 Cl\right]\left(NO_3\right)_2$	q	Diamagentic
c	$\left[Pt\left(en\right)\left(NH_3\right)Cl\right]NO_3$	r	exhibits cis-trans isomerism
d	$\left[Co\left(NH_3\right)_4\left(NO_3\right)_2\right]Br$	s	exhibits ionisation isomerism
		t	Octahedral with d^2sp^3

20. Match the column

	COLUMN-I		COLUMN-II
a)	$Ph-\underset{\underset{O}{\|}}{C}-H \xrightarrow{\text{alkaline KCN}}$ product	**p)**	Final product formed give positive tollen test
b)		**q)**	Final product give test with 2,4DNP
c)	$CH_3-\underset{\underset{O}{\|}}{C}-OEt \xrightarrow[(2)H_3O^{\oplus}\Delta]{(1)C_2H_5ONa}$ Product	**r)**	Final product react with $NaHCO_3$ and liberated CO_2 gas
d)	$Ph-CH=O \xrightarrow[(2)H^{\oplus}]{(1)KOH}$ Product	**s)**	Final product react with Na and Liberated H_2 gas
		t)	Final product lose CO_2 on heating

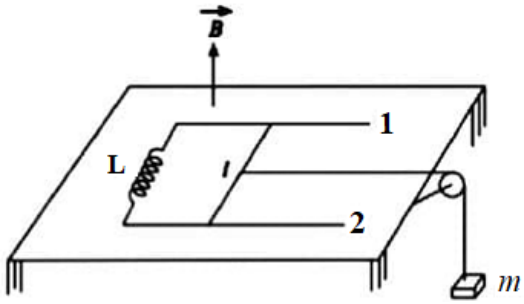
PHYSICS: **Max.Marks: 80**

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

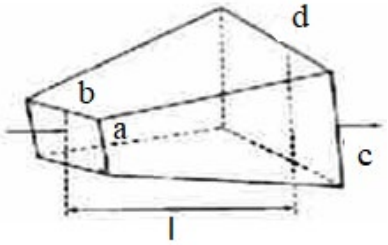
This section contains **8 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

21. A pair of fixed parallel horizontal conducting rails of negligible resistance is connected to an inductance L as shown in the figure. A constant magnetic field B exists perpendicular to the table. The distance between the rail is l. A conducting rod of negligible resistance and mass *m* can slide on the rail's frictionlessly. The rod is connected to mass *m* by a light inextensible non conducting string passing over a light frictionless pulley as shown. Calculate speed of the rod as a function of distance x from initial position.



A) $\sqrt{gx+\frac{B^2l^2x^2}{4mL}}$ **B)** $\sqrt{gx-\frac{B^2l^2x^2}{2mL}}$ **C)** $\sqrt{gx-\frac{B^2l^2x^2}{4mL}}$ **D)** $\sqrt{\frac{gx}{2}-\frac{B^2l^2x^2}{4mL}}$

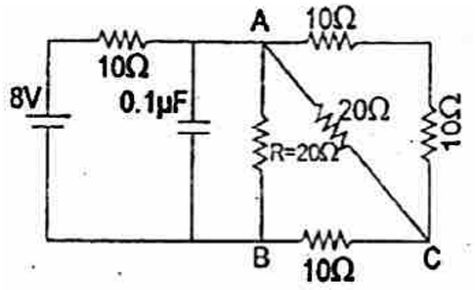
22. A long prismatic conductor or of varying rectangular cross-section and of length l is shown in the figure. The dimension of rectangular cross-section on the one end is $a\times b$ and on the other end it becomes $c\times d$. The side length of rectangular cross-section varies linearly with the distance from one end. The resistivity of the material of the conductor is ρ. Find the resistance of the conductor, if the current enters from the one end and leaves from the other end.



A) $\frac{\rho l}{ad-bc}\ln\left(\frac{ad}{bc}\right)$ **B)** $\frac{\rho l}{a^2-b^2}\ln\left(\frac{d}{c}\right)$

C) $\frac{\rho l}{dc-ab}\ln\left(\frac{cd}{ab}\right)$ **D)** $\frac{\rho l}{d^2-b^2}\ln\left(\frac{a}{c}\right)$

23. A capacitor of capacitance $0.1\mu F$ is connected a battery of emf 8V as shown in the fig. Under steady state condition, choose correct option



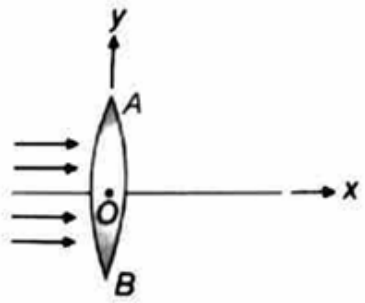
A) Current in the resistor (R) between points A &B is 0.1 A

B) Charge on the capacitor is $0.2\mu C$

C) Charge on the capacitor is $0.4\mu C$

D) Current in the resistor (R) between points A & B is 0.4 A

24. Monochromatic light rays parallel to x-axis strike a thin convex lens AB of refractive index 1.5. If the lens oscillates such that AB tilts up to a small angle θ (in radian) on either side of y-axis, the distance between extreme positions of oscillating image is $f(\sec\theta-\alpha)$. The value of α is



A) $f(\sec\theta-1)$ **B)** $f(2-\sec\theta)$ **C)** $f\left(\frac{1}{\cos\theta}+1\right)$ **D)** $f\left(1-\frac{1}{\cos\theta}\right)$

25. For an isosceles prism of angle A and refractive index μ, it is found that the angle of minimum deviation $\delta_m = A$. Which of the following is incorrect?

A) For the angle of incidence $i_1 = A$, the ray inside the prism is parallel to the base of the prism

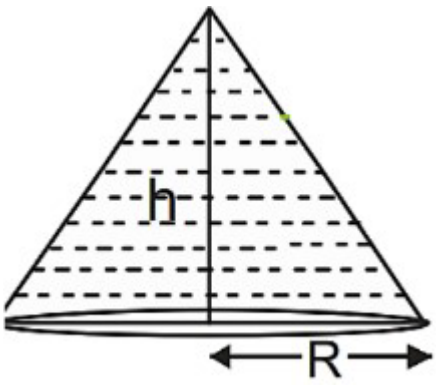
B) For the prism, the refractive index μ and the angle of prism A are related as

$$A=\frac{1}{2}\cos^{-1}\left(\frac{\mu}{2}\right)$$

C) At minimum deviation, the incident angle i_1 and the refracting angle r_1 at the first refracting surface are related by $r_1 = (i_1 / 2)$

D) For this prism, the emergent ray at the second surface will be tangential to the surface when the angle of incidence at the first surface is $i_1 = \sin^{-1}\left[\sin A\sqrt{4\cos^2\frac{A}{2} - 1} - \cos A\right]$

26. A liquid of density ρ is filled in a conical vessel as shown in figure. Force exerted by liquid on side wall is



A) $\frac{2}{3}\pi R^2 \rho g h$ **B)** $\frac{1}{3}\pi R^2 \rho g h$ **C)** $\pi R^2 \rho g h$ **D) 0**

27. The current density in a long wire of radius 'a' varies with radial distance r as $J = kr^2$, where k is a constant.

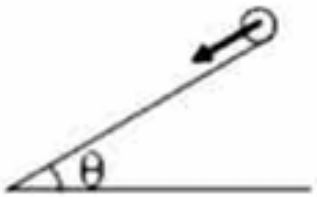
A) Total current passing through the cross section of the wire is $I = \frac{\pi k a^4}{4}$

B) Total current passing through the cross section of the wire is $I = \frac{3\pi k a^3}{2}$

C) The magnetic field at a distance $r > a$ is $B = \frac{\mu_0 k a^4}{2r}$

D) The magnetic field at a distance $r < a$ is $B = \frac{\mu_0 k a^3}{4r}$

28. A solid sphere is in pure rolling motion on an inclined surface having inclination θ



A) Keeping mass and θ constant if radius of solid sphere increases, friction will increase

B) Friction is dissipative force

C) Friction will increase its angular velocity and decreases its linear velocity

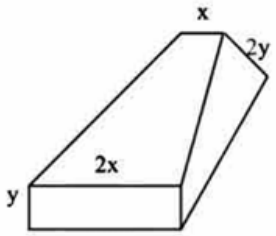
D) If θ decreases, friction will increase

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

29. A conductor is made of an isotropic material (resistivity ρ) has rectangular cross – section. Horizontal dimension of the rectangle decreases linearly from 2x at one end to x at the other end and vertical dimension increases from y to 2y as shown in figure. Length of the conductor along the axis is equal to λ. A battery is connected across the ends of this conductor then



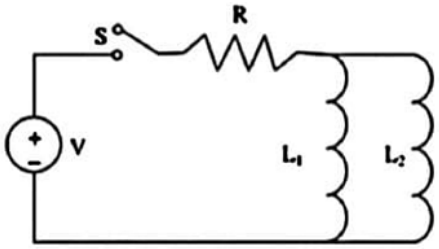
A) Rate of generation of heat per unit length is maximum at the ends of conductor

B) Rate of generation of heat per unit length is maximum at middle cross section

C) Drift velocity of conduction electrons is minimum at middle section

D) At the ends of the conductor, electric field intensity is same

30. A source of constant voltage V is connected to a resistance R and two inductors $L_1 \text{ and } L_2$ through a switch S as shown. There is no mutual inductance between the two inductors. The switch S is initially open. At t=o, the switch is closed and current begins to flow. Which of the following options is/are correct?



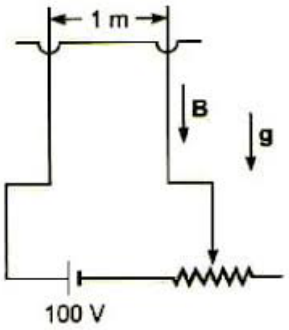
A) After a long time, the current through L_1 will be $\frac{V}{R}\frac{L_2}{L_1+L_2}$

B) After a long time, the current through L_2 will be $\frac{V}{R}\frac{L_1}{L_1+L_2}$

C) The ratio of the currents through $L_1 \text{ and } L_2$ is fixed at all times $(t>0)$

D) At t=0, the current through the resistance R is $\frac{V}{R}$

31. A horizontal rod of mas 2kg is kept touching two vertical parallel rough rails, carrying current. There is a magnetic field B = 2T present vertically downward. The rails are connected to battery of 100 V at t = 0. The resistance of the circuit is 5Ω and starts to increase at constant rate $0.5\Omega / s$. The coefficient of friction between the rails and rod is $\mu = 3/4 (g = 10 m/s^2$ and separation between the rails in 1 m). Then



A) The friction force acting on the rod at t = 0 is 20 N

B) The friction force acting on the rod at t = 0 is 30 N

C) Acceleration of rod at t = 0 is $5m/s^2$

D) At t = 5 sec. rod is about to start

32. Suppose the potential energy between electron and proton at a distance r is given by $\frac{Ke^2}{3r^3}$, application of Bohr's theory to hydrogen atom in this case shows that

A) Energy in the nth orbit is proportional to n^6

B) Energy is proportional to m^{-3} (m = mass of electron)

C) Energy the nth orbit is proportional to n^{-2}

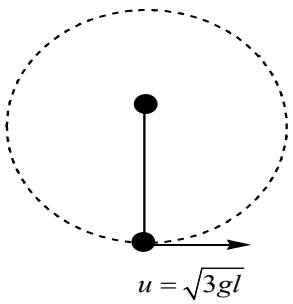
D) Energy is proportional to m^3 (m = mass of electron)

SECTION –III

(INTEGER ANSWER TYPE)

This section contains **6 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS.

33. In the arrangement shown, a bob of mass m is suspended by means of a light and inextensible string of length ℓ connected to a peg P. If the bob is given a horizontal velocity u having magnitude $\sqrt{3g\ell}$, the minimum speed of the bob in subsequent motion is found out to be $\sqrt{\frac{g\ell}{3x}}$ Find the value of x?



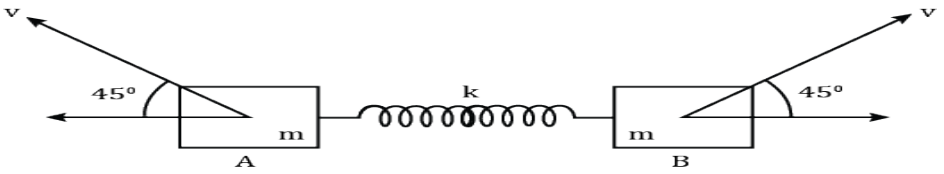
34. A neutron is scattered through (=deviation from its old direction) θ degree in an elastic collision with an initially stationary deuteron. If the neutron loses $\frac{2}{3}$ of its initial K.E, to

the deuteron then find the value of $\frac{\theta}{10}$ (In atomic mass unit, the mass of a neutron is lu and mass of a deuteron is 2u)

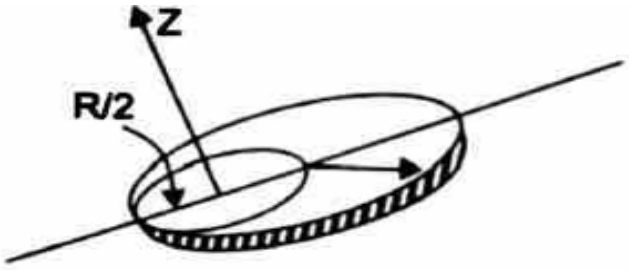
35. A radioactive material consists of nuclides of 3 isotopes which decay by $\alpha-$ emission, $\beta-$ emission, and deuteron emission respectively. Their half-lives are $T_1 = 400\,\text{sec}, T_2 = 800\,\text{sec}\; and\; T_3 = 1600\,\text{sec}$ respectively. At t= 0, probability of getting α, β and deuteron from radio nuclide are equal. If the probability of α emission at t = 1600 seconds is $\frac{n}{13}$, then find the value of 'n' .

36. Two radioactive samples $A_1\; and\; A_2$ having half-life 3 years and 2 years respectively have been decaying for many years. Today the number of atoms in the sample A_1 is twice the number of atoms in the sample A_2. Both the samples had same number of atoms X years ago, calculate X.

37. Blocks A and B of mass m are connected with spring of constant k. Both block lie on frictionless ground and are imparted horizontal velocity v as shown when spring is upstretched. The maximum stretch of spring is $v\sqrt{\frac{xm}{8k}}$. Then find the value of x.



38. Find the moment of inertia(I) of a thin disc of mass m and radius R with hole of radius $\frac{R}{2}$ around the Z-axis (see figure), if m = 6kg, R = 2m (m =6kg) is the mass of the disc after removing the hole. If $I = (20 + x)\,kg\,m^2$, where 'x' is



SECTION –IV

(MATRIX-MATCH TYPE)

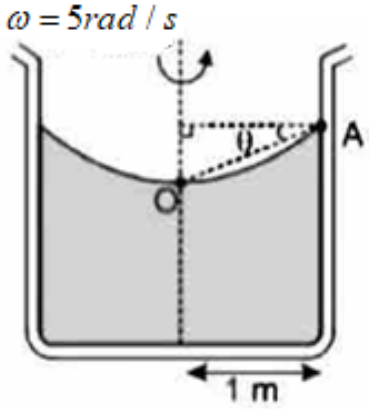
This section contains 2 questions. Each question has **for statements** (A,B,C and D) given in **Column I** and **five statements** (p,q,r,s, and t) in **Column II.** Any given statement in **Column I** can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statements B, darken the bubbles corresponding to q and r in the ORS.

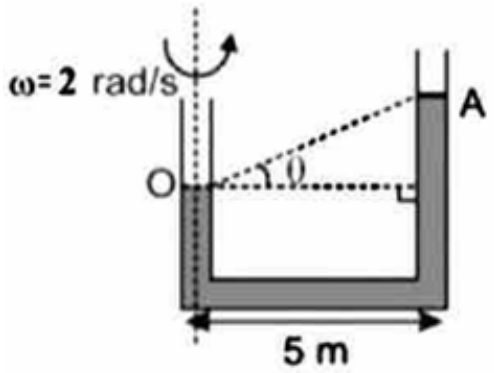
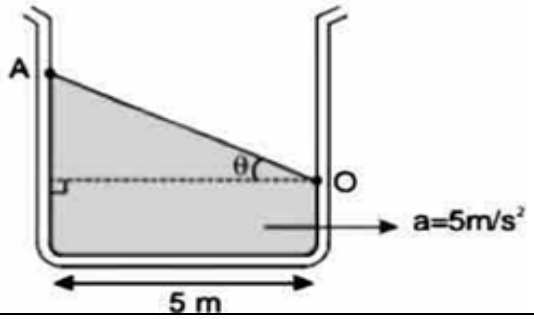
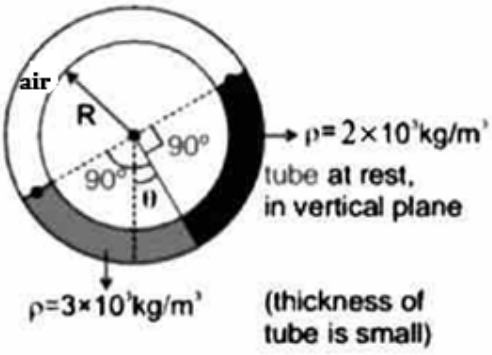
39. A particle enters a space where exists uniform magnetic field $\vec{B} = B_x\vec{i} + B_y\vec{j} + B_z\vec{k}$ and uniform electric field $\vec{E} = E_x\vec{i} + E_y\vec{j} + E_z\vec{k}$ with initial velocity $\vec{u} = u_x\vec{i} + u_y\vec{j} + u_z\vec{k}$

Depending on the values of various components the particle selects a path. Match the entries of column I with the entries of column II. The components other than specified in column I in each entry are non – zero. Neglect gravity.

	COLUMN-I		**COLUMN-II**
A)	$B_y = B_z = E_x = E_z = 0 \quad u = 0$	**P)**	Circle
B)	$E = 0 u_x B_x + u_y B_y \neq -u_z B_z$	**Q)**	Helix with uniform pitch and constant radius
C)	$\vec{u} \times \vec{B} = 0, \vec{u} \times \vec{E} = 0, u \neq 0, B \neq 0, E \neq 0$	**R)**	Cycloid
D)	$\vec{u} \perp \vec{B}, \vec{B} \parallel \vec{E}$	**S)**	Helix with variable pitch and constant radius
		T)	Straight line

40. Match the difference in level and angle (θ) of line connecting O to A in List I with the respective options in list II. (Liquid is ideal in all cases and $g = 10 ms^2$)

List – I		**List – II**	
A)		P)	The difference in the vertical height between O and A is 1.25 m

B)		Q)	The difference in the vertical height between O and A is 5m
C)		R)	$\theta = \tan^{-1}\left(\frac{1}{2}\right)$
D)		S)	$\theta = \tan^{-1}\left(\frac{1}{5}\right)$

MATHEMATICS: **Max.Marks: 80**

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

41. If $\frac{\sin^4\alpha}{a}+\frac{\cos^4\alpha}{b}=\frac{1}{a+b}$, then the value of $\frac{\sin^8\alpha}{a^3}+\frac{\cos^8\alpha}{b^3}$ is

A) $\frac{1}{4(a+b)^3}$ **B)** $\frac{1}{2(a+b)^3}$ **C)** $\frac{1}{(a+b)^3}$ **D)** $\frac{2}{(a+b)^3}$

42. If position vectors of the points A, B and C are a, b, c respectively and the points D and E divide line segments AC and AB in the ratio 2:1 and 1:3, respectively. Then, point of intersection of BD and EC divides EC in the ratio

A) 2:1 **B)** 1:3 **C)** 1:2 **D)** 3:2

43. $\min\left[(x_1-x_2)^2+\left(12-\sqrt{1-x_1^2}-\sqrt{4x_2}\right)^2\right]\forall x_1,x_2\in R$ is:

A) $4\sqrt{5}+1$ **B)** $4\sqrt{5}-1$ **C)** $\sqrt{5}+1$ **D)** $\sqrt{5}-1$

44. The number of positive integral solution of the equation $\begin{vmatrix} x^3+1 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix} = 11$ is

A) 0 **B)** 3 **C)** 6 **D)** 12

45. Let $C_1 \equiv (x-1)^2 + y^2 = 1$ and $C_2 : x^2 + y^2 = r^2$ intersect at P in 1st quadrant and let R be the point where the lines passing through the points $Q \equiv (0, r)$ and P intersects the x-axis. If radius of C_2 decreases indefinitely then limiting position of R is given by $(\alpha, 0)$ then 'α' is equal to

A) 1 **B)** 2 **C)** 3 **D)** 4

46. The maximum value of $F(x) = \int_{7\pi/6}^{x} (a\sin t + b\cos t)\,dt,\ a, b, \in R^+, x \in \left[\frac{7\pi}{6}, \sqrt{21}\right]$ is

A) a+b **B)** $a\sqrt{21} - b\frac{7\pi}{6}$ **C)** 0 **D)** None of these

47. If any point O inside or outside a tetrahedron ABCD is joined to the vertices and AO, BO, CO, DO are produced so as to cut the planes of the opposite faces in P,Q,R,S respectively, the value of $\frac{OP}{AP} + \frac{OQ}{BQ} + \frac{OR}{CR} + \frac{OS}{DS}$ is

A) $\frac{1}{2}$ **B)** 1 **C)** 2 **D)** 4

48. Given $f(x) = \begin{cases} \log_a \left(a\left|[x] + [-x]\right|\right)^x \left(\dfrac{a^{\left(\frac{[x]+[-x]}{|x|}\right)^{\frac{2}{-5}}}}{3 + a^{\frac{1}{|x|}}} \right); & |x| \neq 0; a > 1 \\ 0 \quad ; & x = 0 \end{cases}$

Where [] represents the integral part function, then:

A) f is continuous but not differentiable at x=0

B) f is cont. and diff. at x=0

C) The differentiability of 'f' at x=0 depends on the value of a

D) f is continuous and differentiable at x=0 and for a=e only

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

49. If $f:R\to[a,6]$ defined by $f(x)=\dfrac{x^2-2x+d}{x^2+3x+d}$ is an onto function. then

A) a is equal to $\dfrac{2}{7}$ **B)** d is equal to 4

C) a is equal to 1 **D)** d-7a is equal to-3

50. $\displaystyle\int_{-\infty}^{\infty}\frac{\ln|t|}{x^4+t^2}dt-\frac{\pi}{2}\ln 2=0$, then

A) Number of values of x satisfying the equation is 5

B) Roots of $x^2-\left(2+\sqrt{2}\right)x+2\sqrt{2}=0$ satisfy the above equation

C) Product of values of x satisfying above equation is 8

D) Roots of $x^3+31x^2-2x-21=0$ satisfy the above equation

51. A ray M is sent along the line $\dfrac{x-0}{2}=\dfrac{y-2}{2}=\dfrac{z-1}{0}$ and is reflected by the plane x=0 at point A. The reflected ray is again reflected by the plane $x+2y=0$ at point B. The initial ray and final reflected ray and final reflected ray meets at point J. Then:

A) The co-ordinates of point B is $(4,-2,1)$

B) The co-ordinates of point J is $(-3,-1,1)$

C) The centroid of ΔABJ is $(0,0,0)$

D) The co-ordinates of point J is $(2,-1,1)$

52. If $a+b=k$, when a, b>0 and $S(k,n)=\displaystyle\sum_{r=0}^{n}r^2\left({}^nC_r\right)a^r.b^{n-r}$, then

A) $S(1,3)=3\left(3a^2+ab\right)$ **B)** $S(2,4)=16\left(4a^2+ab\right)$

C) $S(3,5)=25\left(5a^2+ab\right)$ **D)** $S(4,6)=36\left(6a^2+ab\right)$

SECTION –III

(INTEGER ANSWER TYPE)

This section contains **6 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS

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53. Let $z=\left(\cos 12^{o}+i\sin 12^{o}+\cos 48^{o}+i\sin 48^{o}\right)^{6}$, then Im(z) is equal to _____

54. A conference attended by 200 delegates is held in a hall. The hall has 7doors, marked A, B, ……, G. At each door, an entry book is kept and the delegates entering through that door sign it in the order in which they enter. If each delegate is free to enter any time and through any door he likes, if the total no. of different sets of seven lists would arise in all is equal to ${}^{n}P_{r}$ then $'n-r'$is equal to (assume that every person signs only at his first entry)

55. The number of point where $\left|xf\left(x\right)\right|+\left\|x-2\right|-1\right|$ is non-differentiable in $x\in\left(0,3\pi\right)$, where $f\left(x\right)=\prod_{k=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^{k}}\right)}{3}\right)$ is_____

56. A bag contained 3 maths book 2 physics books. A book is drawn at random. If it is of math, 2 more books of maths together with this book put back in the bag and if it is of physics it is not replaced in the bag. This experiment is replaced 3 time. If third draw given math book, the probability that first two drawn books were of physics is $\frac{p}{q}$

(where H.C.F (p, q) =1) then q-8(p+1) =_____

57. Consider the bisector of the angle between the lines $x+2y-11=0\,and\,3x-6y-5=0$ which contains the point $\left(1,-3\right)$. Let $\left(a,1\right)and\left(b,2\right)$ be two points lying on this line. The value of $a-b$ is

58. The plane denoted by $P_{1}:4x+7y+4z+81=0$ is rotated through a right angle about its line of intersection with the plane $P_{2}:5x+3y+10z=25$. If the plane in its new position be denoted by P, and the distance of this plane from the origin is d, then the value of $\left[\frac{k}{2}\right]$.(where[.] represents greatest integer less than or equal to k) is_____

SECTION –IV

(MATRIX-MATCH TYPE)

This section contains 2 questions. Each question has **for statements** (A,B,C and D) given in **Column I** and **five statements** (p,q,r,s, and t) in **Column II.** Any given statement in **Column I** can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statements B, darken the bubbles corresponding to q and r in the ORS.

59. let f(x) be a real valued function defined by $f(x) = x^2 - 2|x|$ *and*

$$g(x) = \begin{cases} \text{minimum}\{f(t) : -2 \le t \le x\}, & x \in [-2, 0) \\ \text{maximum}\{f(t) : 0 \le t \le x\}, & x \in [0, 3] \end{cases}$$

	COLUMN-I		COLUMN-II
A)	f(x) is not differentiable at x equal to	**P)**	-2
B)	g(x) is not differentiable but continuous at x equal to	**Q)**	0
C)	Number of points of non- differentiability of g(x) is equal to	**R)**	1
D)	Absolute maximum value of g(x) is equal to	**S)**	2
		T)	3

60. C_1 is a curve which is obtained by taking image of $\sin^{-1} x$ *w.r.t* the curve $\cos^{-1} x$ along y-axis and C_2 is a curve which is obtained by taking image of $\tan^{-1} x$ w.r.t $\cot^{-1} x$ along y-axis. Then based on the following grid, give your answer (where [.] represents greatest integer function)

	COLUMN-I		COLUMN-II
A)	If maximum value of C_2 *is* α then $[\alpha]$ is equal to	**P)**	0
B)	If minimum value of C_1 *is* β then $[\beta]$ is equal to	**Q)**	7
C)	If C_1 *and* C_2 intersect at $x = \gamma$, then $[\gamma]$ is equal to	**R)**	4
D)	If curve $C_1 + C_2$ takes δ integral values, then δ is equal to	**S)**	-2
		T)	14